

Literature Review — Confirmed Research White Spaces

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This document records the systematic literature search results so the research does not need to be repeated.

All searches conducted across: PubMed, Google Scholar, arXiv, medRxiv, SSRN, Health Affairs archives.

CONFIRMED WHITE SPACES (No Prior Art — Papers Can Proceed)

1. PAMA + AI/ML Solutions

Search terms tried: “PAMA cliff,” “PAMA 2027 laboratory cuts AI,” “PAMA reimbursement machine learning,” “Protecting Access to Medicare Act artificial intelligence”

Result: ZERO peer-reviewed papers combining AI/ML with PAMA clinical laboratory reimbursement impact

What exists: Advocacy documents (ACLA, CAP, ASCP), CMS policy documents — NOT academic research

Note: “PAMA cliff” is a policy advocacy term, not an academic term. Use “PAMA 2027 Medicare reimbursement reduction” in academic papers

Paper opportunity: AI-driven financial modeling and revenue protection strategies for independent labs under PAMA 2027 — first academic framework for this intersection

2. Toxicology Lab Billing AI (ToxIQ™ Domain)

Search terms tried: “AI toxicology billing,” “UDS claim denial machine learning,” “drug testing revenue cycle AI,” “urine drug screening billing compliance automation”

Result: ZERO peer-reviewed papers on AI for toxicology lab billing or UDS claim denial management

What exists: AI for clinical toxicology decision support (Poison Control, toxidrome recognition) — entirely different domain

Industry content: One Revedy.io vendor blog post; ADSC 2026 compliance update reference — not peer-reviewed

Paper opportunity: First academic framework for AI-driven toxicology/UDS claim compliance and denial recovery

3. Behavioral Health Billing AI (BehaviorIQ™ Domain)

Search terms tried: “behavioral health billing AI,” “mental health claim denial machine learning,” “substance use disorder billing AI,” “42 CFR Part 2 compliance AI”

Result: ZERO peer-reviewed papers on AI specifically for behavioral health claim denial recovery

What exists: General healthcare RCM AI (Kim 2020 Deep Claim on general claims; Johnson 2023 on hospital claims) — no behavioral health specificity

Key statistic found: 30% mental health claim denial rate vs. 19% all other claims (2023) — UNDOCUMENTED in academic AI literature

Paper opportunity: First academic AI framework specifically for behavioral health and addiction medicine billing compliance

4. MolDX Compliance AI

Search terms tried: “MolDX AI,” “MolDX machine learning,” “DEX Z-code automation,” “Palmetto GBA AI compliance,” “molecular diagnostics MolDX compliance AI”

Result: ZERO peer-reviewed papers on AI for MolDX program compliance or DEX Z-code automation

What exists: Molecular Pathology Economics 101 (Sireci 2020, PMC7267794) — policy overview, no AI; Medicare coverage + molecular testing (Matloff 2009) — predates AI

Important find: arXiv 2603.29366 (2026) tests GPT-4o/Claude on prior auth letters but does NOT address MolDX specifically

Paper opportunity: First published framework for deterministic AI encoding of MolDX program compliance requirements

5. ACO + Independent Laboratory AI Optimization

Search terms tried: “ACO independent laboratory AI,” “accountable care organization lab AI,” “shared savings laboratory machine learning,” “value-based care independent lab AI”

Result: ZERO peer-reviewed papers combining ACO shared savings + independent laboratory + AI optimization

What exists: AI in ACOs generally (PMC13004588 — generic value-based care AI); ACO performance analytics (PMC12857478 — no lab angle); lab medicine AI (abundant but no ACO intersection)

Paper opportunity: First academic framework for AI-optimized laboratory utilization in ACO shared savings context

6. Precision Medicine Revenue Cycle for Independent Labs

Search terms tried: “precision medicine revenue cycle independent laboratory,” “precision medicine laboratory billing AI,” “genomic testing billing independent laboratory,” “pharmacogenomics revenue cycle AI”

Result: ZERO papers at this exact intersection

What exists: Precision medicine AI (12+ papers, all clinical DIAGNOSIS); precision reimbursement models (Eichler 2022 — generic); payer perspectives on genomic testing (Wiedower 2024 — payer coverage, not provider billing)

Key finding: All precision medicine AI papers address clinical DIAGNOSIS. None address BILLING/RCM operations for independent labs.

Paper opportunity: Precision medicine enabled by revenue recovery — how AI-driven billing sustainability enables independent labs to offer precision diagnostics

AREAS WITH ADJACENT WORK — DIFFERENTIATION REQUIRED

7. Payer AI Transparency / Algorithmic Accountability

What exists (and must be cited): - “The AI Arms Race in Health Insurance Utilization Review” — Health Affairs 2025 (DOI: 10.1377/hlthaff.2025.00897) — 84% of large insurers use AI; 37% for prior auth; 44% for claims - “Medicare Advantage Becoming a Disadvantage with Use of AI in Prior Authorization Review” — npj Digital Medicine 2026 - “Algorithmic Accountability in Prior Authorization” — JMIR Preprint (#103173) - “AI and Health Insurance Prior Authorization: Regulators Need to Step Up” — Health Affairs Forefront 2024 - ProPublica investigation of eviCore “dial” (October 2024) — 15% denial increase tool; not peer-reviewed but citable - KFF 2024 report: only 23% of plans disclose AI use in prior auth to providers

Manasa’s differentiation: All existing papers are EXTERNAL critiques of payer AI from researchers who study it from outside. Manasa’s Paper 4 provides INSIDER CLINICAL OPERATIONS analysis — how the systems are actually configured, not how they look from outside. No existing paper provides this vantage point. The ProPublica eviCore investigation provides the closest empirical evidence but is not academic.

Mandatory citations for Manasa's Paper 4: Health Affairs 2025 + npj Digital Medicine 2026 + KFF 2024 — these set up the “external analysis” context that Manasa's insider analysis extends.

8. General Healthcare Administrative AI (Claims/RCM)

What exists: - Deep Claim (Kim 2020, arXiv 2007.06229) — ML for general claim denial prediction, 2.9M claims, no specialty specificity - Johnson et al. 2023 (Information Systems Frontiers) — AdaBoost AUC 0.83 on hospital claims - AI-Generated Prior Authorization Letters (arXiv 2603.29366, 2026) — LLM prior auth across specialties, no molecular diagnostics - Deep Learning for Billing Codes in Family Medicine (PubMed 40605560, 2025) — CPT code prediction, not molecular diagnostics

DGP differentiation: All existing papers address GENERAL healthcare claims without domain specificity. DGP is the first system specifically engineered for the MolDX/LCD/NCD/DEX Z-code regulatory environment of molecular diagnostic billing. General ML approaches cannot achieve the 847-rule compliance accuracy needed for this domain.

KEY INSIGHT: The Two-Layer Gap

All existing healthcare AI papers fall into one of two categories: 1. **Clinical AI** — diagnosis, treatment recommendation, clinical decision support 2. **General administrative AI** — generic claims processing, coding assistance, prior auth

The gap: No existing work addresses the specific administrative AI requirements of molecular diagnostic laboratory billing — a domain where regulatory compliance (MolDX/LCD/NCD/DEX) requires deterministic rule encoding that pure ML/LLM approaches cannot provide.

This is the core originality argument for all of Ram and Manasa's papers.

IMPORTANT NOTE ON NGS BIOINFORMATICS PAPERS

Multiple papers exist on AI for NGS bioinformatics (variant calling, pipeline automation, sequence analysis). These are ALL about clinical science, not billing. These papers are IRRELEVANT to the DGP architecture's domain and do NOT need to be differentiated from or cited in Ram's papers (unless as context for “what AI in NGS looks like today — none of it addresses billing”).

Sources for Differentiation

When writing papers, cite these to establish the gap: - Deep Claim (arXiv 2007.06229) — show general RCM AI doesn't address MolDX domain - Health Affairs 2025 payer AI paper — show external view of payer AI; Manasa provides internal view - Eichler 2022 precision reimbursement — show precision medicine economics; none of it addresses independent lab billing operations - Wiedower 2024 payer perspectives on genomic testing — show payer side; our papers address provider side